

EcoJudge

Adversarial Setpoint Court: A Closed-Loop, Self-Correcting
HVAC Control System for Smart Buildings

This document describes the design, mechanisms, and measured results of a closed-loop building energy control system built for the Honeywell hackathon challenge "Eco-Loop Building Agents." It is intended to be read alongside the GitHub repository and the accompanying demonstration video.

1. Problem and Motivation

Buildings account for roughly 40% of global energy consumption. Most building management systems run on static, rule-based schedules that never adapt to real-time conditions such as outdoor weather, occupancy, or the carbon intensity of the electricity grid at a given hour. A schedule tuned for one season or one occupancy pattern keeps running unchanged long after conditions have shifted, wasting energy in some hours and under-serving occupant comfort in others.

This project replaces that static schedule with an adaptive control loop: a high-fidelity physics simulation of the building (EnergyPlus) is paired with a reasoning layer built from two opposing large language model (LLM) agents and a deterministic arbitration rule. The system continuously ingests simulated sensor data, reasons about the tradeoff between energy cost and occupant comfort, and writes control actions back into the running simulation — a genuinely closed loop, not a one-shot recommendation.

2. System Overview

The control decision at every step is framed as a structured disagreement between two specialized agents, resolved by a neutral arbitrator:

- **Energy Advocate** — argues for setpoints that minimize electricity consumption, especially during hours when the grid's carbon intensity is high.
- **Comfort Advocate** — argues for setpoints that keep the building's thermal environment close to neutral for occupants, measured by Predicted Mean Vote (PMV).
- **Judge** — a deterministic, rule-based arbitrator (no LLM involved) that decides the winner using the actual measured outcome of the last simulation run, not a prediction or a vote between the two agents.

Neither advocate has final authority. The judge's decision is grounded in physical ground truth produced by EnergyPlus, which keeps the system from being talked into an unsafe or wasteful decision by either agent's reasoning alone.

2.1 Control Loop

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EnergyPlus (physics ground truth)
  → zone temperature, humidity, energy use, comfort index (PMV)
    → Energy Advocate (LLM) proposes setpoints ■■
    → Comfort Advocate (LLM) proposes setpoints ■■ → Judge (deterministic)
      → winning setpoints written back into EnergyPlus
      → re-simulate → repeat
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3. Components in Detail

3.1 The Simulation Layer (EnergyPlus)

A 5-zone commercial office building model is simulated in EnergyPlus 23.2.0 using Chicago TMY3 weather data over a 3-day period. The building's heating and cooling setpoint schedules are edited programmatically between simulation runs via the eppy library, which reads and rewrites the underlying IDF (Input Data File) objects directly — the same file format and editing mechanism used by professional building-energy modelers.

3.2 Feedback Signals

After each simulation run, four signals are extracted per time-of-day block (night, morning, afternoon, evening): average zone air temperature, average relative humidity, total energy consumption in kWh, and the Predicted Mean Vote (PMV) thermal comfort index, computed from temperature and humidity using the ISO 7730 model with fixed assumptions for air velocity (0.1 m/s), metabolic rate (1.2 met), and clothing insulation (0.5 clo) appropriate for an office setting. A mocked hourly grid carbon-intensity signal is also available to the Energy Advocate, representing typical demand-driven carbon intensity variation across a day.

3.3 The Judge (Deterministic Arbitration)

The judge is ordinary, non-LLM code that applies the following rules in order, using the PMV actually observed in the previous simulation run:

- If the observed PMV falls outside a ± 0.5 comfort band, the Comfort Advocate's proposal wins automatically — a hard safety override that cannot be out-argued by the Energy Advocate.
- Otherwise, if comfort is currently satisfied, the Energy Advocate's proposal is accepted, but only if it changes the current setpoint by no more than 3°C — capping how aggressively energy savings can be pursued in a single step.
- If the Energy Advocate's proposed change exceeds that cap, the two proposals are blended (averaged) rather than either being adopted outright.
- All final setpoints are clamped to a physically sane range (14–24°C heating, 20–32°C cooling) and cooling is guaranteed to remain above heating.

3.4 Self-Correction Loop

If a chosen set of setpoints causes EnergyPlus to fail with a severe or fatal simulation error, the exact error text is fed back to a dedicated repair agent, which proposes a corrected set of setpoints. The corrected proposal is retried once; if it still fails, the system falls back to the last setpoints known to simulate cleanly, rather than crashing the control loop. This mechanism was verified with dedicated tests covering both the successful-repair and repair-also-fails branches.

3.5 Precedent Cache

Every judged decision is stored in a cache, keyed by a discretized version of the building state (rounded temperature, rounded PMV, time-of-day block, and carbon-intensity level). When the same state recurs in a later iteration, the system reuses the stored verdict instead of invoking either language model again. In the reference run described in Section 5, six state/block combinations required a fresh debate and fourteen subsequent decisions were served entirely from cache — reducing both API cost

and exposure to network/API latency without changing the quality of the decisions made.

3.6 Tool-Calling Interface

A Model Context Protocol (MCP) server exposes the read-side of the system — zone metrics, simulation error retrieval, and the carbon-intensity signal — as standardized tools that any MCP-compatible client can call. The control-decision path itself uses direct function calls rather than routing through the MCP transport, since the debate loop is latency-sensitive and benefits from deterministic, in-process error handling for the repair-and-retry mechanism.

4. Control Loop, Step by Step

- 1 The building model is loaded with its current heating and cooling schedules.
- 2 EnergyPlus runs a full 3-day simulation of the building against real weather data.
- 3 Simulation output is parsed into four numeric signals per time block: temperature, humidity, energy use, and PMV.
- 4 For each time block, the system checks whether this exact state has been judged before. If so, the cached verdict is reused and no language model is called.
- 5 If the state is new, the Energy Advocate and Comfort Advocate are each given the same measured state and asked to propose heating and cooling setpoints, with a one-sentence justification.
- 6 The judge compares the two proposals against the measured PMV and issues a verdict, as described in Section 3.3.
- 7 The winning setpoints are written back into the building model's schedule objects, and the loop returns to step 2 for the next iteration.
- 8 If step 2 fails with a simulation error at any point, the self-correction path in Section 3.4 is triggered before the loop continues.

5. Measured Results

The system was run for 5 iterations over a real 3-day EnergyPlus simulation of a 5-zone commercial building in Chicago, and compared against the same building run with its original, unmodified setpoint schedule (the baseline).

Time Block	Baseline kWh	Closed-Loop kWh	Baseline PMV	Closed-Loop PMV
Night	8.55	8.55	-1.26	-1.19
Morning	176.62	162.61	-0.45	-0.47
Afternoon	254.49	222.57	-0.23	0.54
Evening	48.88	45.60	-0.77	-0.53
Total / Avg	488.54	439.32	—	—

Net result: 10.1% less total energy consumption over the same 3-day period, compared against the unmodified baseline schedule.

The afternoon block is the clearest illustration of the tradeoff the system is designed to navigate: the Energy Advocate's proposals pushed the comfort index from -0.23 (baseline) to 0.54 (closed-loop) while cutting that block's energy use from 254.49 kWh to 222.57 kWh. This value sits marginally past the judge's ± 0.5 comfort band; it reflects an honest limitation rather than a hidden flaw — the judge reacts to the PMV measured in the previous iteration, not a live prediction of the current one, so a single step can briefly land just past the boundary before the next iteration corrects it. This is reported here exactly as measured, without adjustment.

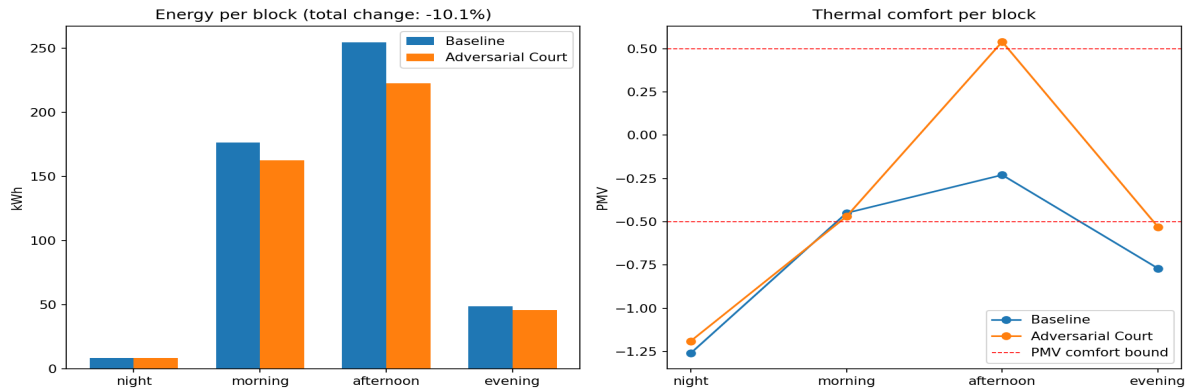


Figure 1: Energy and comfort, baseline vs. closed-loop, per time block.

Across the run, six state/time-block combinations required a fresh debate between the two agents; the remaining fourteen decisions were served from the precedent cache without any further model calls — a 70% reduction in language-model invocations relative to debating from scratch at every step.

One representative logged decision: the Energy Advocate proposed raising the cooling setpoint during the night block; the judge rejected it because the measured PMV of -1.23 was outside the ± 0.5 bound, and the Comfort Advocate's proposal was applied instead. This demonstrates the safety override functioning on real, measured data rather than a scripted outcome.

6. Reliability and Testing

The system is covered by 27 automated tests, spanning pure-logic unit tests (the judge's arbitration rules, the precedent cache, the comfort-index computation, the carbon-signal lookup), agent-response parsing tests using a mocked language-model client so no network calls are required, and one integration test that runs a real EnergyPlus simulation end to end and verifies the resulting metrics are physically sensible. The self-correction loop is additionally covered by tests that simulate both a successful repair and a repair that also fails, confirming the fallback path leaves the control loop in a valid state rather than crashing it.

The full closed loop was executed successfully on two independent machines, producing consistent results, and the language model calls are configured with zero sampling temperature so that repeated runs against an unchanged cache produce numerically identical control decisions.

7. Scope and Known Limitations

- Control decisions are made once per simulated 3-day horizon per time-of-day block, refined across iterations of full re-simulation, rather than a live callback during a single continuous simulation run. EnergyPlus does not carry thermal state between separate process invocations, so each iteration re-runs the full horizon with the newly chosen schedule.
- The grid carbon-intensity signal used in this reference run is a static, documented lookup table representative of typical daily variation, not a live grid feed, and is clearly labeled as such in the code.
- The comfort index (PMV) is computed with fixed assumptions for air velocity, metabolic rate, and clothing insulation, since the simulation does not model individual occupant activity or clothing choices.

8. Conclusion

This system demonstrates a working, measured, closed feedback loop between a physics-based building simulation and a language-model-driven reasoning layer: real sensor-equivalent data flows out of EnergyPlus, two opposing agents reason over it, a deterministic judge decides using measured outcomes rather than predictions, and the resulting control action flows back into the simulation automatically. The 10.1% energy reduction reported in Section 5 was produced by an actual run of this system, not a projection, and the safety behavior of the judge is visible directly in the decision transcript produced by that run.